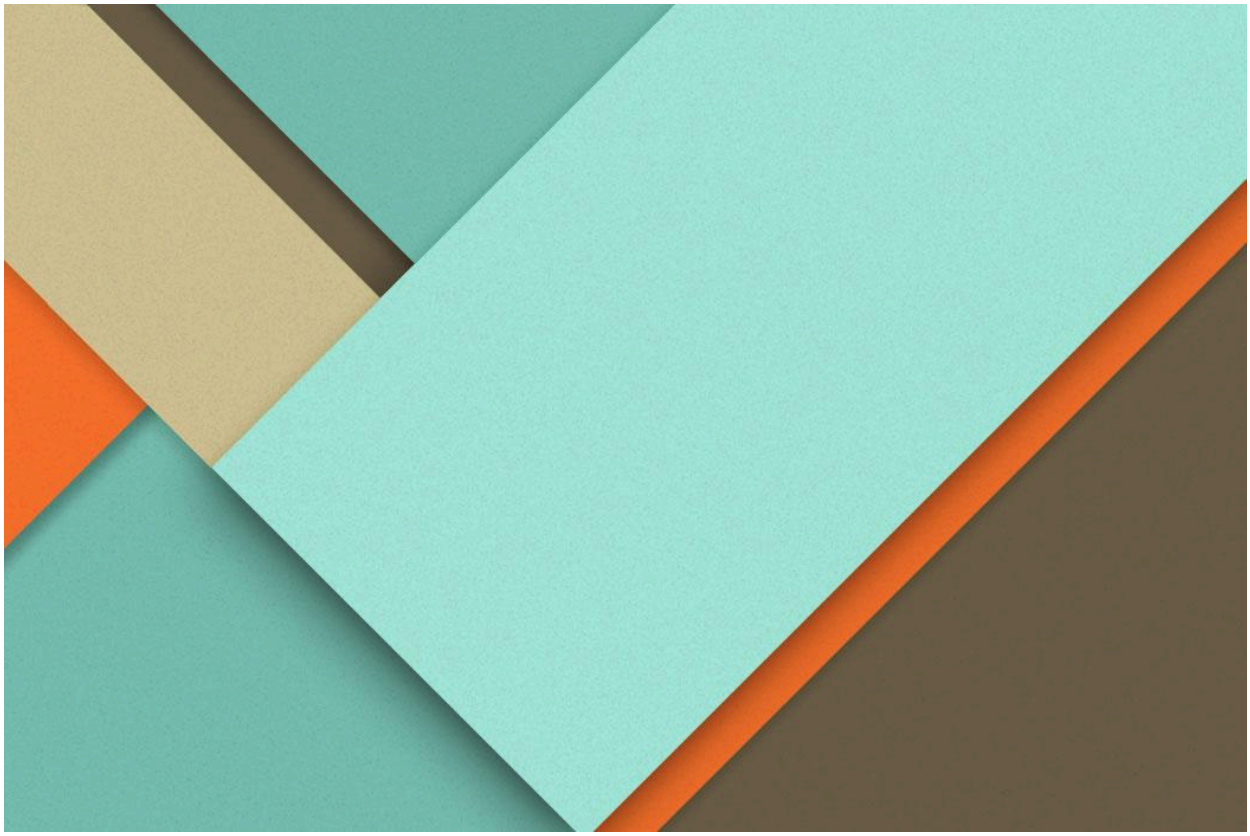


Fixed Income Securities

QFGB-8931-001

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Part II. TIPS

1. Methodology

1.1 Overview of the Arbitrage Strategy

The replicating strategy involves three positions. First, the arbitrageur purchases a TIPS issue with the same maturity as the target Treasury bond. The TIPS pays inflation-adjusted coupons of sI_t each period and a principal of $100I_t$ at maturity, where s is the TIPS coupon rate and I_t is the CPI-U inflation index ratio at date t relative to the issuance date. Second, for each coupon date, the arbitrageur enters a zero-coupon inflation swap with notional s (or s+100 at the final date). The fixed leg of each swap pays $s(1+f_t)^t$ at date t, where f_t is the inflation swap rate for maturity t. The floating leg pays sI_t, which exactly cancels the inflation component of the TIPS coupon, converting it into a fixed payment of $s \cdot F_t$ where $F_t = (1+f_t)^t$. Third, small positions in Treasury STRIPS are taken at each coupon date to bridge the gap between $s \cdot F_t$ and the target coupon c.

The net result is a portfolio that delivers exactly the same cash flows as the nominal Treasury bond at every date, at a total cost equal to the TIPS price V plus the cost of the STRIPS positions. The mispricing is the difference between the Treasury bond's full price P and this synthetic price.

1.2 Key Formulas

The core formulas used throughout the analysis are as follows:

Fixed leg of inflation swap at date t:

$$\mathbf{F_t = (1 + f_t)^t}$$

STRIPS notional required at each coupon date (periods 1 to T−1):

$$\mathbf{x_t = c - s \cdot F_t}$$

STRIPS notional at the final date T (coupon + principal):

$$\mathbf{x_t = (c + 100) - (s + 100) \cdot F_t}$$

Synthetic price of the replicating portfolio:

$$\textbf{Synthetic Price} = V + \sum x_i \cdot D(t)$$

where D(t) is the price of a Treasury STRIP maturing at date t, expressed per $1 face value (Bloomberg quotes STRIPS per $100 face; we divide by 100 to match the paper's definition). The inflation swaps carry no upfront cost under standard swap conventions.

1.3 Deviations from the Paper's Methodology

Three simplifications were made compared to the paper. First, the paper uses cubic spline interpolation across swap maturities of 1 through 30 years. We use linear interpolation between adjacent integer tenors (USSWIT1–USSWIT7), taking the simple average of the two surrounding rates for each half-year maturity. The impact is minimal given the flat swap curve observed in March 2026 (range of approximately 28 basis points across the full curve). Second, the paper adjusts for maturity mismatches by computing the YTM of the synthetic bond and repricing it to exactly match the Treasury maturity. We use the raw synthetic price directly. For the 31-day mismatch on a seven-year bond the impact is negligible.

2. Data

2.1 Bond Pair Selection

Following FLL, we select a TIPS issue with a maturity date within 31 days of the target Treasury bond. The selected pair is the 3.500% Treasury bond maturing February 15, 2033 and the 1.125% TIPS maturing January 15, 2033. The maturity mismatch is exactly 31 days, the maximum permitted under the paper's sample construction rules.

Security	Maturity	Coupon	Clean Price	Full Price
Treasury Bond	2/15/2033	3.500%	96-19+ (96.6094)	$96.9284
TIPS	1/15/2033	1.125%	97-15¼ (97.4766)	$97.7384

2.2 Price Conversion

Treasury and TIPS prices are quoted in 32nds notation on Bloomberg. The conversions are: Treasury clean price 96-19+ = 96 + 19.5/32 = 96.609375; TIPS clean price 97-15¼ = 97 + 15.25/32 = 97.476563. Accrued interest is computed from the Bloomberg-reported dollar amount on a $1,000,000 face value position and converted to per $100 notional by dividing by 10,000. Treasury accrued: $3,190.61 / 10,000 = $0.3191; TIPS accrued: $2,618.64 / 10,000 = $0.2619. Both bonds had 64 days of accrued interest as of settlement.

2.3 Inflation Swap Rates and Interpolation

Zero-coupon USD inflation swap rates were obtained from Bloomberg for integer maturities 1 through 7 years. Half-year maturities are obtained by linear interpolation between adjacent tenors. For the 0.5-year period, the 1-year rate is used directly as an extrapolation, consistent with the paper's treatment of sub-one-year maturities.

Tenor	Ticker	Market Rate	Period (yrs)	Interpolated Rate	Calculation	F□
1Y	USSWIT1	2.704%	0.5	2.704%	Flat extrapolation from 1Y	1.01087
1Y	USSWIT1	2.704%	1.0	2.704%	Market Quote	1.02704
2Y	USSWIT2	2.522%	1.5	2.613%	(2.704+2.522)/2	1.03934
2Y	USSWIT2	2.522%	2.0	2.522%	Market Quote	1.05105
3Y	USSWIT3	2.477%	2.5	2.500%	(2.522+2.477)/2	1.06355
3Y	USSWIT3	2.477%	3.0	2.477%	Market Quote	1.07591
4Y	USSWIT4	2.451%	3.5	2.464%	(2.477+2.451)/2	1.08980
4Y	USSWIT4	2.451%	4.0	2.451%	Market Quote	1.10310
5Y	USSWIT5	2.420%	4.5	2.436%	(2.451+2.420)/2	1.11520
5Y	USSWIT5	2.420%	5.0	2.420%	Market Quote	1.12732

6Y	USSWIT6	2.444%	5.5	2.432%	(2.420+2.444)/2	1.14207
6Y	USSWIT6	2.444%	6.0	2.444%	Market Quote	1.15740
7Y	USSWIT7	2.425%	6.5	2.435%	(2.444+2.425)/2	1.17113
7Y	USSWIT7	2.425%	7.0	2.425%	Market Quote	1.18316

2.4 Treasury STRIPS Prices

Treasury STRIPS prices were obtained from Bloomberg as ask prices per $100 face value. As discussed in the methodology, these are divided by 100 to produce the per-dollar discount factor D(t) used in the paper's formulas. The 14 remaining coupon dates from settlement through maturity are shown below.

#	Date	STRIPS Price	D(t)	#	Date	STRIPS Price	D(t)
1	8/15/2026	98.5150	0.985150	8	2/15/2030	86.1220	0.861220
2	2/15/2027	96.8006	0.968006	9	8/15/2030	84.3968	0.843968
3	8/15/2027	94.8820	0.948820	10	2/15/2031	82.6731	0.826731
4	2/15/2028	93.0997	0.930997	11	8/15/2031	80.9573	0.809573
5	8/15/2028	91.3442	0.913442	12	2/15/2032	79.1574	0.791574
6	2/15/2029	89.6130	0.896130	13	8/15/2032	77.4806	0.774806
7	8/15/2029	87.8689	0.878689	14	2/15/2033	75.6917	0.756917

Date	Treasury CF	TIPS CF	Inflation Swap CF	STRIPS CF x□	STRIPS Price D(t)	Total CF
0	**−$96.9284**	**−$97.7384**	**$0**	**−$0.2251**	**—**	**−$97.9635**
8/15/2026	1.7500	$0.5625 \cdot I_1$	$0.5625 \cdot (F_1 - I_1)$	1.18862	98.5150	**1.7500**
2/15/2027	1.7500	$0.5625 \cdot I_2$	$0.5625 \cdot (F_2 - I_2)$	1.18245	96.8006	**1.7500**

8/15/2027	1.7500	$0.5625 \cdot I_3$	$0.5625 \cdot (F_3 - I_3)$	1.17637	94.8820	**1.7500**
2/15/2028	1.7500	$0.5625 \cdot I_4$	$0.5625 \cdot (F_4 - I_4)$	1.17093	93.0997	**1.7500**
8/15/2028	1.7500	$0.5625 \cdot I_5$	$0.5625 \cdot (F_5 - I_5)$	1.16513	91.3442	**1.7500**
2/15/2029	1.7500	$0.5625 \cdot I_6$	$0.5625 \cdot (F_6 - I_6)$	1.15926	89.6130	**1.7500**
8/15/2029	1.7500	$0.5625 \cdot I_7$	$0.5625 \cdot (F_7 - I_7)$	1.15192	87.8689	**1.7500**
2/15/2030	1.7500	$0.5625 \cdot I_8$	$0.5625 \cdot (F_8 - I_8)$	1.14731	86.1220	**1.7500**
8/15/2030	1.7500	$0.5625 \cdot I_9$	$0.5625 \cdot (F_9 - I_9)$	1.14169	84.3968	**1.7500**
2/15/2031	1.7500	$0.5625 \cdot I_{10}$	$0.5625 \cdot (F_{10} - I_{10})$	1.13521	82.6731	**1.7500**
8/15/2031	1.7500	$0.5625 \cdot I_{11}$	$0.5625 \cdot (F_{11} - I_{11})$	1.12912	80.9573	**1.7500**
2/15/2032	1.7500	$0.5625 \cdot I_{12}$	$0.5625 \cdot (F_{12} - I_{12})$	1.12288	79.1574	**1.7500**
8/15/2032	1.7500	$0.5625 \cdot I_{13}$	$0.5625 \cdot (F_{13} - I_{13})$	1.11669	77.4806	**1.7500**
2/15/2033	**101.7500**	**$100.5625 \cdot I_{14}$**	**$100.5625 \cdot (F_{14} - I_{14})$**	**−16.9362**	**75.6917**	**101.7500**

3. Mispricing Analysis

3.1 Results

The mispricing calculation is summarised in Table 5:

Item	Value (per $100)	Notes
Treasury Bond Full Price (P)	$96.9284	
TIPS Full Price (V)	$97.7384	
Total STRIPS Cost (∑x□·D(t))	$0.2251	Sum of 14 positions; net long
Synthetic Price (V + STRIPS Cost)	$97.9635	Cost of replicating portfolio
Mispricing (P − Synthetic)	**−$1.0351**	Negative → Treasury is cheap relative to TIPS

3.2 Interpretation

The mispricing of −$1.035 per $100 notional indicates that the synthetic replicating portfolio is more expensive than the actual Treasury bond by approximately $1.04. In other words, TIPS are relatively expensive compared to nominal Treasury bonds in March 2026. This is the opposite direction from the mispricing documented by FLL over their 2004–2009 sample, where Treasury bonds were almost uniformly overvalued relative to TIPS.